

Practice Worksheet

Answer from memory before consulting the flash cards. Use code examples where requested.

Name: _____ Date: _____

Directions: Write the answer in your own words. After checking your work, mark each item as **New**, **Review**, or **Mastered**.

HTML Foundations

1. Draw the basic HTML5 document structure and label the purpose of head and body.

2. What is semantic HTML, and how does it support assistive technology?

3. Explain the difference between a form's action and method attributes.

4. Write a labeled required email input and explain how the label connects to it.

CSS Foundations

Use diagrams when they make the relationship easier to see.

5. Draw and label the four layers of the CSS box model.

6. Explain why box-sizing: border-box is useful on modern websites.

7. Draw the main axis and cross axis for flex-direction: row. Where do justify-content and align-items operate?

8. Rewrite a fixed-width card so it can fit a 320px mobile viewport.

9. What does a media query do? Write one that changes a layout below 768px.

JavaScript Data Flow

Focus on what each operation returns and whether it mutates the original value.

10. Explain primitive versus reference values with one object example.

11. Write a map expression that changes an array of objects into an array of names.

12. Write a filter expression that keeps only active items.

13. Show an immutable way to append an item to an existing array.

14. What is a shallow copy? What nested data can it still share?

15. Why is using forEach when you need a new array a mistake?

Async JavaScript and Network Debugging

Trace the request from the browser to the data displayed on screen.

16. Draw the `fetch` lifecycle from pending Promise to parsed JSON.

17. Why must you check `response.ok` manually for HTTP errors?

18. List five details you would record while inspecting a failed request in DevTools.

19. Write the four visible states a data-fetching interface should handle.

20. Correct this bug: `const data = fetch('/api/items');`

React and Next.js Architecture

Explain the direction of data and the reason behind each state decision.

21. Draw unidirectional data flow between a parent component and a child component.

22. When should a React state update use an updater function?

23. Why should derived values usually not be stored in separate state variables?

24. Why do list items need stable unique keys? Why can an array index be risky?

25. What can a Server Component not use? How does use client change the boundary?

Review, Debugging, and Self-Assessment

Use precise evidence before asking an AI assistant for help.

26. Write a four-part AI prompt: context, data model, action/state, and layout constraints.

27. List the three steps in systematic error triage.

28. Name three red flags you would look for in AI-generated React or CSS.

29. What evidence will you save each day and each Friday?

Confidence Check

HTML: New / Review / Mastered	CSS: New / Review / Mastered
JavaScript: New / Review / Mastered	React: New / Review / Mastered
Debugging: New / Review / Mastered	AI review: New / Review / Mastered